FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVII

JEE (Main)-2021

TEST DATE: 14-08-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

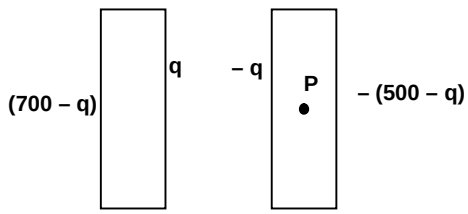
Sol. Since charge on cavity = 0. Therefore whole charge transfers to outermost surface and electric field will be present in the region,
$r > R_2$ only

2. D

Sol. Charge acquired by the plates of the capacitor.

$q_0 = CV = (10\mu F)(50V) = 500\mu C$

Now let the charge distribution is as follows



Total charge on positive plate has now become 700 μC while that in negative plate is still – 500 μC

Here, charges are in μC.

Net electric field at P is zero.

$$\therefore \frac{(700-q)}{2A\varepsilon_0}+\frac{q}{2A\varepsilon_0}+\left(\frac{500-q}{2A\varepsilon_0}\right)=\frac{q}{2A\varepsilon_0}$$

$\therefore \quad q = 600\mu C$

$\therefore$ Potential difference between the plates is

$$\Delta V = \frac{q}{C} = \frac{600\mu C}{10\mu F} = 60V$$

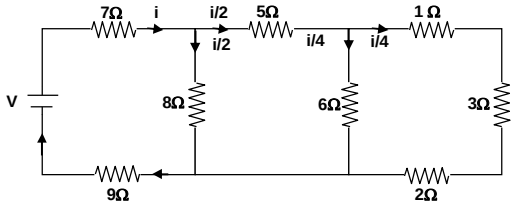
3. B

Sol. $R_{net} = 20\Omega$

$\therefore$ current drawn from the circuit is

$$i = \frac{V}{20}$$

Distribution of this current *i* is as shown below



Given $\frac{i}{4} = 0.25$

or $\frac{(V/20)}{4} = 0.25$

$\therefore$ V = 20 V

4. D

Sol. Battery is short circuited. Therefore, potential difference across its terminals will be zero.

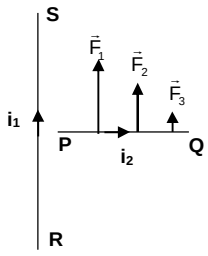
5. C

Sol. $p = \frac{2\pi m}{Bq}(v\cos 45^0) = \frac{2\pi m}{Bq}(v\sin 45^0)$

$\therefore$ $\frac{mv\sin 45^0}{Bq} = \frac{p}{2\pi}$ = radius of helix

6. C

Sol. Since, the magnetic field due to wire RS is non uniform. Neither the force nor the torque on wire PQ is zero. Therefore, it will have both translational as well as rotational motion.



7. B

Sol. During entering and during exiting time upto when flux is changing is:

$$t = \frac{a}{v}$$

$\therefore$ Total time $= 2t = \frac{2a}{v}$

8. D

Sol. $Z = \sqrt{(3)^2 + (4)^2} = 5\Omega$

$i_0 = \frac{10}{Z} = 2A$

$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53^0$

$\therefore \quad i = 2\sin\left(\omega t - 53^0\right)$

At $\quad t = \dfrac{\pi}{\omega}$ or $\omega t = \pi$

$i = 2\sin\left(\pi - 53^0\right)$

$= 2 \times \dfrac{4}{5} = 1.6A$

$\therefore \quad V_R = iR = 4.8V$

9. A

Sol. $\dfrac{1}{v} - \dfrac{3/2}{+30} = \dfrac{1-3/2}{+20}$

$\therefore \quad v = 40\,cm$

10. B

Sol. Deviation by a sphere is $(i-r)$

Here, deviation $\delta = 60^0 = 2(i-r)$

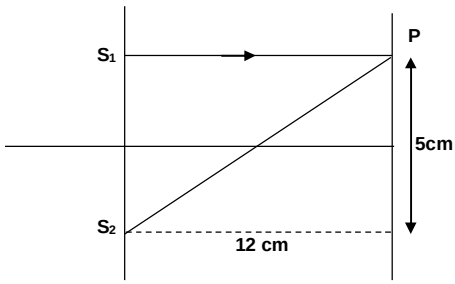
or $\quad i - r = 30^0$

$\therefore \quad r = i - 30^0 = 60^0 - 30^0 = 30^0$

$\therefore \quad \mu = \dfrac{\sin i}{\sin r} = \dfrac{\sin 60^0}{\sin 30^0} = \sqrt{3}$

11. A

Sol. $\Delta x = S_2P - S_1P$



$= \sqrt{(12)^2 + (5)^2} - 12$

$= 1\,cm$

For minimum intensity,

$\Delta x = \dfrac{\lambda}{2}, \dfrac{3\lambda}{2}, \dfrac{5\lambda}{2}$ .....etc.

12. C

Sol. AB process is isochoric

$\therefore \quad Q_1 = nC_V\Delta T$

$= n\left(\dfrac{5}{2}R\right)\left(T_B - T_A\right)$

$= 2.5\left(nRT_B - nRT_A\right)$

$= 2.5\left(P_BV_B - P_AV_A\right)$

$= 2.5P_0V_0$

BC process is isothermal.

$$\therefore \quad Q_2 = nRT_B \ln\left(\frac{P_i}{P_f}\right)$$

$$= P_B V_B \ln 2$$

$$= (2P_0V_0)(0.7)$$

$$= 1.4P_0V_0$$

$$Q_{net} = Q_1 + Q_2 = 3.9P_0V_0$$

13. A

Sol. $$H_1 + H_2 = \frac{KA(T_1 - T_2)}{3\ell} + \frac{KA(T_1 - T_2)}{\ell} = \frac{4}{3\ell}KA(T_1 - T_2)$$

In later case

$$H_2 = 2H - H_1 = \frac{7KA}{3\ell}(T_1 - T_2) = \frac{K'A}{\ell}(T_1 - T_2)$$

$$\Rightarrow \quad K' = \frac{7}{3}K$$

14. A

Sol. $$\Delta\ell = \frac{F\ell}{AY} \qquad (T = F)$$

$$= \frac{T\ell}{AY}$$

$$\therefore \quad T = \frac{YA\Delta\ell}{\ell}$$

Lowest frequency $= \frac{v}{2\ell} = \sqrt{\frac{T/\rho A}{2\ell}}$

$$f = \frac{1}{2\ell}\sqrt{\frac{YA\Delta\ell}{\rho/A}}$$

$$= \frac{1}{2\ell}\sqrt{\frac{YA\Delta\ell}{\rho\ell}}$$

Substituting the values,

$$f = \frac{1}{2\times 1}\sqrt{\frac{9\times 10^{10}\times 0.05\times 10^{-2}}{(9000)(1)}}$$

= 35 Hz

15. C

Sol. $$f' = \frac{9}{8}f = f\left(\frac{340 + V}{340 - V}\right)$$

Solving this equation we get,

$v = 20\,m/s$

16. B

Sol. Mean position will be at kx = mg

or $$x = \frac{mg}{k} = \frac{8\times 10}{200} = \frac{2}{5}m = 0.4m$$

This is also the amplitude of oscillation

i.e., $A = 0.4m$

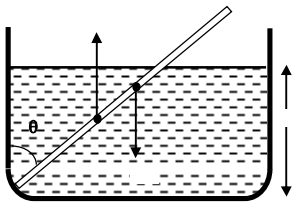
Now, $v_{max} = A\omega = A\sqrt{\frac{k}{m}}$

$= (0.4)\sqrt{\frac{200}{8}} = 2\ m/s$

17. C

Sol. Length of rod inside the water

$= 1.0 \sec\theta = \sec\theta$



Upthrust $F = \left(\frac{2}{2}\right)(\sec\theta)\left(\frac{1}{500}\right)(1000)(10)$

or $F = 20 \sec\theta$

Weight of rod $W = 2 \times 10 = 20\ N$

For rotational equilibrium of rod net torque about O should be zero.

$\therefore \quad F\left(\frac{\sec\theta}{2}\right)(\sin\theta) = W(10\sin\theta)$

or $\frac{20}{2}\sec^2\theta = 20$

or $\theta = 45^0$

$\therefore \quad F = 20\sec 45^0$

$= 20\sqrt{2}\,N$

For vertical equilibrium of rod. Force exerted by the hinge on the rod will be $\left(20\sqrt{2} - 20\right)N$ downwards or 8.28 N ~ 8.3 N downwards.

18. B

Sol. In crease in length due to own weight

$I = \int_0^L \frac{(L-x)mg\,dx}{LAY} = \frac{mgL}{2AY}$ (Here, $m = \lambda L$)

or $Y = \frac{mgL}{2Al} = \frac{\lambda gL^2}{2Al}$

19. C

Sol. At all places potential will increase by $\frac{GM}{R}$.

20. B

Sol. $W = 5\ eV - 2eV = 3ev$

In second case,

$K_{max} = E - W$

$= 6eV - 3eV$

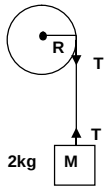
$= 3eV$

Therefore stopping potential is 3V.

SECTION – B

21. 00002.00

Sol. $TR = I\alpha,\ TR = I\dfrac{a}{R}$ or $T = \dfrac{Ia}{R^2}$



$$Mg - T = Ma$$

or $$Mg = Ma + T = \left(M + \frac{I}{R^2}\right)a$$

or $$a = \frac{Mg}{\left(M + \frac{I}{R^2}\right)}$$

$$= \frac{2\times 10}{\left(2 + \frac{0.32}{0.04}\right)} = 2 m/s^2$$

22. 00008.00

Sol. $K = \sqrt{\dfrac{\ell}{m}}$ or $K \propto \sqrt{\ell}$

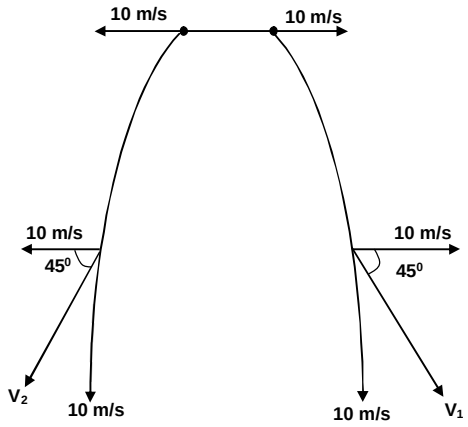
$$\therefore \quad \frac{K_1}{K_2} = \sqrt{\frac{\ell_1}{\ell_2}} \Rightarrow \frac{10}{n} = \sqrt{\frac{I_C + m(6)^2}{I_C}}$$

$$\Rightarrow \quad \frac{10}{n} = \sqrt{1 + \frac{36m}{mn^2}} = \sqrt{1 + \frac{36}{n^2}}$$

Solving this equation we get n = 8

23. 00001.00

Sol.



24. 00000.09

Sol. $f_r = mR\omega^2$

$$= mR(2\pi f)^2 = 4\pi^2 f^2\, mR$$

$$= 4\left(\frac{22}{7}\right)^2\left(\frac{2}{\sqrt{10}}\right)^2\left(5\times 10^{-3}\right)\left(5\times 10^{-2}\right)$$

$$= \left(9\times 10^{-3}\right)N$$

25. 00007.00

Sol. Decrease in potential energy = work done against resistance for

$\therefore \quad mg(h+x) = Fx$

$\therefore \quad F = \dfrac{mg(h+x)}{x} = \dfrac{0.2\times 10\times 7}{2}$

$= 7N$

26. 00024.00

Sol. When force is applied on A

$$\frac{F_A}{m_A+m_B} = \frac{\mu m_A g}{m_B} \qquad \ldots(i)$$

When force is applied on B

$$\frac{F_B}{m_A+m_B} = \frac{\mu m_A g}{m_A} \qquad \ldots(ii)$$

Dividing these two equations, we get

$$\frac{F_B}{F_A} = \frac{m_B}{m_A}$$

$\therefore \quad F_B = \dfrac{m_B}{m_A}.F_A = \dfrac{8}{4}\times 112 = 24N$

27. 00040.00

Sol. Standard equation of projectile motion

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

Comparing with given equation

$$A = \tan\theta \text{ and } B = \frac{g}{2u^2\cos^2\theta}$$

So, $\dfrac{A}{B} = \dfrac{\tan\theta\times 2u^2\cos^2\theta}{g} = 40$

$(As\,\theta = 45^0,\ u = 20m/s,\ g = 10m/s^2)$

28. 00008.00

Sol. Area of a-t graph gives change in velocity. When net area will become zero, particle will acquire its initial velocity.

29. 00000.04

Sol. $eV = \dfrac{hc}{\lambda} - W$

$\therefore \quad V = \left(\dfrac{h}{e}\right).\dfrac{c}{\lambda} - \dfrac{W}{e}$

$$V_1 = \left(\frac{h}{e}\right)\frac{c}{\lambda_1} - \frac{W}{e} \qquad \ldots(i)$$

$$V_2 = \left(\frac{h}{e}\right)\frac{c}{\lambda_2} - \frac{W}{e} \qquad \ldots(ii)$$

Solving these two equations, we get,

$$\frac{h}{e}=\frac{\lambda_1\lambda_2\left(V_1-V_2\right)}{c\left(\lambda_2-\lambda_1\right)}=\frac{\left(0.6\times0.4\times10^{-12}\right)\left(1.0\right)}{\left(3\times10^{8}\right)\left(02\times10^{-6}\right)}$$

$$=4\times10^{-15}\,V-s$$

30. 00003.46

Sol. $$i_{rms}^2=\frac{\int i^2dt}{\int dt}$$

$$=\frac{\int_2^4(4t)dt}{\int_2^4 dt}=\frac{4\int_2^4 t\,dt}{2}=2\left[\frac{t^2}{2}\right]_2^4$$

$$=12A^2$$

$$i_{rms}=2\sqrt{3}\,A$$

Chemistry

PART – B

SECTION – A

31. A

Sol. Isotones have same number of neutrons. All atoms in triad (a) have same number of neutrons $(= A - Z = 8)$.

32. C

Sol. Ionization potential is amount of energy required to take out most loosely bonded electron from neutral atom. Its value depends on stability of atom.
(electronic configuration)

$C - 1s^2 2s^2 2p^2$ $C^+ - 1s^2 2s^2 2p^1$

$N - 1s^2 2s^2 2p^3$ $N^+ - 1s^2 2s^2 2p^2$

$O - 1s^2 2s^2 2p^4$ $O^+ - 1s^2 2s^2 2p^3$

$F - 1s^2 2s^2 2p^5$ $F^+ - 1s^2 2s^2 2p^4$

As for IE_2 the electron in all the cases is to be removed from 2p orbital so it must follow the order $C < N < O < F$ (i.e. increase across a period)

But in case of O^+, the 2p orbital is half-filled and is more stable as compared to others. So the order becomes: $C < N < F < O$ or $O > F > N > C$

33. A

Sol. $O_2^- \left(17e^-\right) - \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \left\{\pi 2p_x^2 = \pi 2p_y^2\right\}\left\{\pi^* 2p_x^2 = \pi^* 2p_y^1\right\}$

Thus, O_2^- has one unpaired electron; hence it is paramagnetic. Other species have no unpaired electron. All of them have 14 electrons.

34. A

Sol. The ions of group II of salt analysis are precipitated by HCl and H_2S whereas members of group IV are precipitated by H_2S in alkaline medium.
$\because Bi^{3+}$ and Sn^{4+} both belong to group II
$\therefore$ They will be precipitated by HCl in presence of H_2S.

35. B

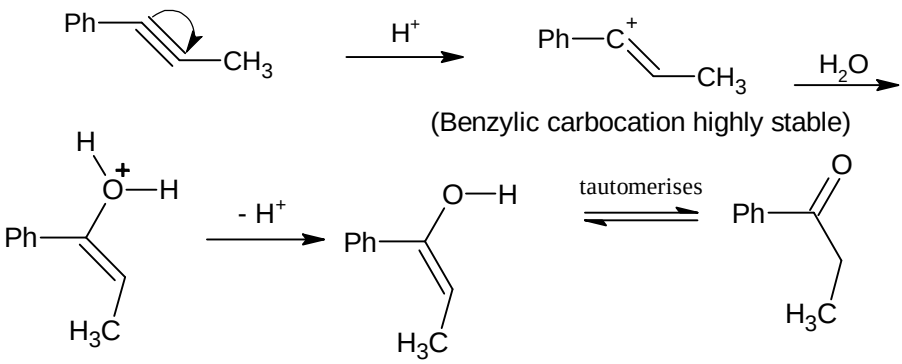
Sol. Volume strength = Normality $\times 5.6 = 1.5 \times 5.6 = 8.4$ L

36. B

Sol. Dipole moment of p-dichlorobenzene is zero because of symmetrical structure. o- and m – dichlorobenzene have higher dipole moments than toluene due to high electronegativity of chlorine than – CH_3 group. Further, the o-dichlorobenzene has higher dipole moment due to lower bond angle than the m-isomer. Hence, the order of increasing dipole moment is:
p-dichlorobenzene (IV) < toluene (I)
< m-dichlorobenzene (II) < o-dichlorobenzene (III)

37. A

Sol.

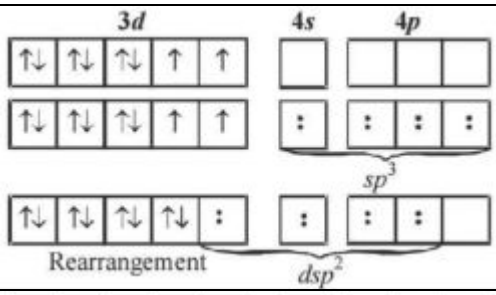
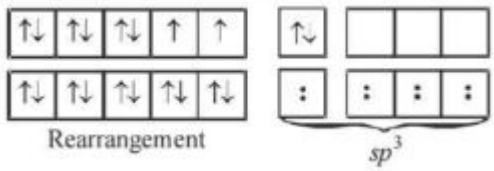


38. A

Sol. Only nitrates of heavy metals and lithium decompose on heating to produce NO_2.

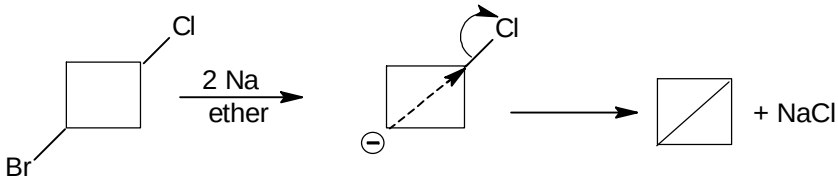
39. C

Sol.

Atom/Ion Complex	Configuration	No. of unpaired electrons	Magnetic nature
$Ni^{2+}\left(3d^8\right)$		2	Paramagnetic
$\left[NiCl_4\right]^{2-}$		2	Paramagnetic
$\left[Ni(CN)_4\right]^{2-}$		0	Diamagnetic
$Ni\left(3d^8 4s^2\right)$		2	Paramagnetic
$\left[Ni(CO)_4\right]$		0	Diamagnetic

40. D

Sol. It is an example of intramolecular Wurtz reaction.



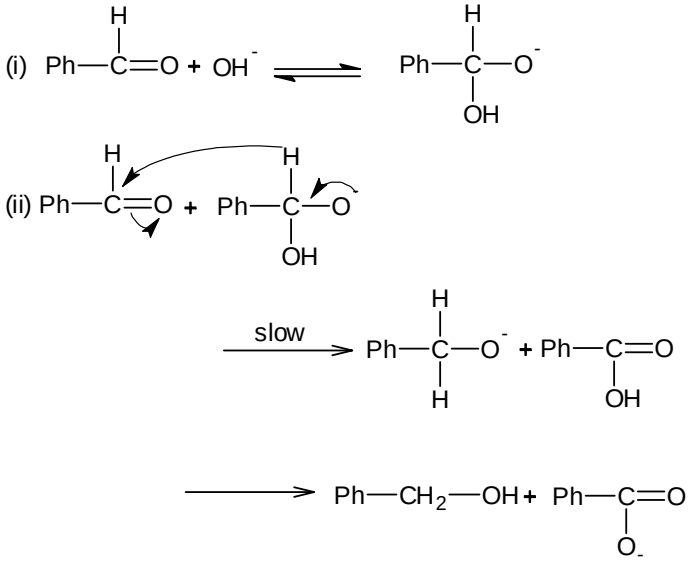
Br^- is a better leaving group than chloride. In this reaction alkali metal (Na) is electron donor.

41. D

Sol. In dye test, phenolic – OH group is converted to $-O^-$ which activates the ring towards electrophilic aromatic substitution

42. B

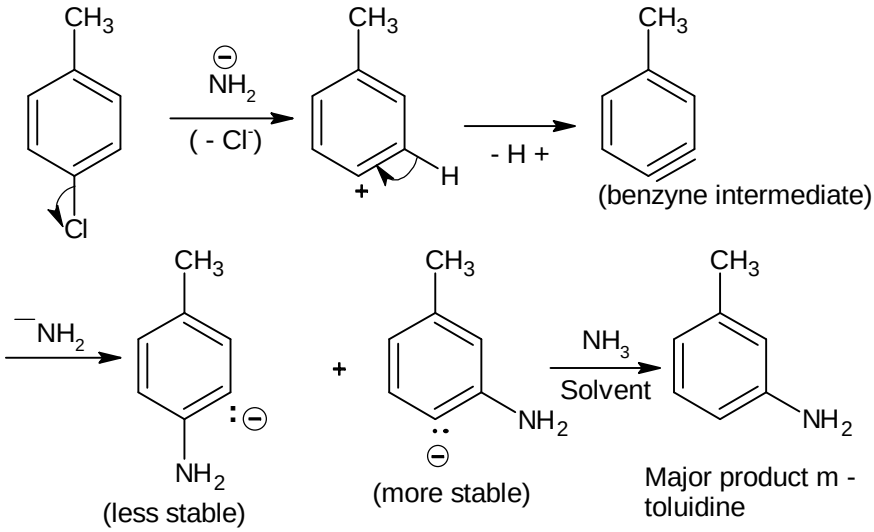
Sol. The mechanism is



The slowest step is the transfer of hydride to the carbonyl group as shown in step (ii).

43. B

Sol.



(due to +I effect of methyl group)
(since hyperconjugation is not applicable on carbanion)

44. D

Sol. Nylon and cellulose, both have hydrogen bonding, polyvinyl chloride has dipole-dipole interaction, while natural rubber has van der Waal forces which are weakest.

45. C

Sol. $\left(P+\frac{a}{V^2}\right)(V-b)=RT$; Here $\left(P+\frac{a}{V^2}\right)$ accounts for the intermolecular forces.

46. A

Sol. Heat capacity at constant volume $\left(q_v\right)=\Delta E$

Heat capacity of constant pressure $\left(q_p\right) = \Delta H$

$\Delta H = \Delta E + \Delta nRT$ or $\Delta H - \Delta E = \Delta nRT$

$\Delta n =$ no. of moles of gaseous products – no. of moles of gaseous reactants

$= 12 - 15 = -3$

$\Delta H - \Delta E = -3 \times 8.314 \times 298 J = -7.43$ kJ.

47. D

Sol. The pH of the solution at the equivalence point will be greater than 7 due to salt hydrolysis. So an indicator giving colour in basic medium will be suitable.
Phenolphthalein is a good indicator if the base is strong because strong base immediately changes the pH at end point.

48. B

Sol. The species $_{13}Al^{29}$ (No. of neutrons = 16) contains more neutrons than the stable isotope $_{13}Al^{27}$ (No, of neutrons = 14). Neutron on decomposition shows $\beta -$ emission.

$_0n^1 \rightarrow {}_{+1}p^1 + {}_{-1}e^0$

β-particle

49. A

Sol. Added HgI_2 forms a complex with KI in the solution as follows

$2KI + HgI_2 \rightarrow K_2\left[HgI_4\right]$

As a result, number of particles decreases and so ΔT_f decreases.

Note Depression in freezing point is a colligative property]

50. A

Sol. $Fe(s) \rightarrow Fe^{2+} + 2e^-; \qquad E^o = 0.44V$

$2H^+ + 2e^- + 1/2O_2 \rightarrow H_2O(\ell); \qquad E^o = +1.23V$

$Fe(s) + 2H^+ + 1/2O_2 \rightarrow Fe^{2+} + H_2O;$

$E^o_{cell} = 0.44 + 1.23 = 1.67V$

$\therefore \Delta G^o = -nFE^o_{cell} = -2 \times 96500 \times 1.67 = -322$ kJ mol^{-1}

SECTION – B

51. 00080.09

Sol. Dipole moment, $\mu = q \times d$ coulombs meter

For KCl $d = 2.6 \times 10^{-10}$ m

For complete separation of unit charge (electronic charge)

$q = e = 1.602 \times 10^{-19}$ C

Hence $\mu = 1.602 \times 10^{-19} \times 2.6 \times 10^{-10} = 4.1652 \times 10^{-29}$ Cm

$\mu_{KCl} = 3.336 \times 10^{-29}$ Coulomb meter (given)

$\therefore$ % Ionic character of KCl $= \dfrac{3.336 \times 10^{-29}}{4.1652 \times 10^{-29}} \times 100$

= 80.09%

52. 00041.33

Sol. Given, moles $= \dfrac{\text{mass}}{\text{mol.wt}}$

mass of gas = 3.7g, mass of hydrogen = 0.184 g

$T_2 = 25\,^oC = 273 + 25K = 298K, T_1 = 17\,^oC = 273 + 17 = 290K$

Moles of $H_2 = n_1 = \dfrac{\text{Mass}}{\text{M. wt.}} = \dfrac{0.184}{2} = 0.092$

Moles of gas $= n_2 = \dfrac{\text{Mass}}{\text{M.wt.}} = \dfrac{3.7}{M}$

For hydrogen $P_1V_1 = n_1RT_1$ ….(i)

For gas $P_1V_1 = n_2RT_2$ ….(ii)

(∵ Pressure and volume of both the gases are same)

∴ From equation (i) and equation (ii)

$\dfrac{P_1V_1}{P_1V_1} = \dfrac{n_1RT_1}{n_2RT_2}$ or $1 = \dfrac{0.092 \times 290}{n_2 \times 298}$

or $n_2 = \dfrac{0.092 \times 290}{298}$ or $\dfrac{3.7}{M} = \dfrac{0.092 \times 290}{298}$

$\therefore M = \dfrac{3.7 \times 298}{0.092 \times 290} = 41.33$

53. 00002.20

Sol. Since the pressure of both the gases are different, and the temperature is constant, the rate at which molecules of the two gas diffuse is directly proportional to the pressure. This rate of diffusion is also directly proportional to the distance travelled by the gas. Hence.

r_1 (of HCl gas) at pressure $P \propto 60 \propto \dfrac{P}{\sqrt{36.5}}$ …(i)

and r_2 (of NH_3) at 1atm $\propto 40 \propto \dfrac{1}{\sqrt{17}}$ …(ii)

From (i) and (ii)

$\dfrac{r_1}{r_2} = \dfrac{60}{40} = \dfrac{P}{\sqrt{36.5}} \times \dfrac{\sqrt{17}}{1}$

$P = \dfrac{60}{40} \times \dfrac{\sqrt{36.5}}{\sqrt{17}} = 2.198\,\text{atm} \approx 2.20\text{atm}$

54. 00309.17

Sol. Given $S(s) + 3F_2(g) \rightarrow SF_6(g); \Delta H = -1100\ \text{kJ mol}^{-1}$ …(i)

$S(s) \rightarrow S(g); \Delta H = 275\text{kJ mol}^{-1}$ …(ii)

$1/\ 2F_2(g) \rightarrow F(g); \Delta H = 80\text{kJ mol}^{-1}$ …(iii)

To get $SF_6(g) \rightarrow S(g) + 6F(g)$ we can proceed as

$(ii) + 6 \times (iii) - (i)$

$\therefore SF_6(g) \rightarrow S(g) + 6F(g); \Delta H = 1855\ \text{kJ mol}^{-1}$

Thus average bond energy for S–F bond $= \dfrac{1855}{6} = 309.17\text{kJ mol}^{-1}$

55. 00078.37

Sol. Volume of blood = 10 ml. (given)

$[H_2CO_3]$ in blood = 2 M (given)

$[NaHCO_3]$ to be added = 5 m (given)

Let volume of $NaHCO_3$ added in 10 ml blood = V ml

$\therefore [H_2CO_3]$ in blood mixture $= \dfrac{2\times 10}{(V+10)}$

$[NaHCO_3]$ in blood mixture $= \dfrac{5\times V}{(V+10)}$

$\therefore pH = pK_a + \log\dfrac{[Salt]}{[Acid]}$

or $7.4 = -\log 7.8\times10^{-7} + \log\dfrac{5V/(V+10)}{20V/(V+10)} \therefore V = 78.37\,ml$

56. 00001.26

Sol. Density in fcc $= \dfrac{n_1\times at.\ wt.}{V_1\times N_o}$

Density in bcc $= \dfrac{n_2\times at.wt.}{V_2\times N_o}$

fcc unit cell length $3.5\,\mathring{A}$; bcc unit cell length = $3.0\,\mathring{A}$

Density in fcc $= \dfrac{n_1\times at.wt.}{V_1\times N_o}$

Density in bcc $= \dfrac{n_2\times at.\ wt.}{V_2\times N_o}$

$\therefore \dfrac{D_{fcc}}{D_{bcc}} = \dfrac{n_1}{n_2}\times\dfrac{V_2}{V_1}$

n_1 for fcc = 4; Also $V_1 = a^3 = \left(3.5\times10^{-8}\right)^3$

n_2 for fcc = 2; Also $V_2 = a^3 = \left(3.0\times10^{-8}\right)^3$

$\therefore \dfrac{D_{fcc}}{D_{bcc}} = \dfrac{4\times\left(3.0\times10^{-8}\right)^3}{2\times\left(3.5\times10^{-8}\right)^3} = 1.259 \approx 1.26$

57. 00023.44

Sol. $\Delta T_t = K_f.m$

$\dfrac{p^o - p}{p^0} = \dfrac{\text{moles of solute}}{\text{moles of solvent}}$

Depression in freezing point, $\Delta T_f = 0-(-0.30) = 0.30$

Now we know that $\Delta T_f = K_f m$

$\therefore m = \dfrac{\Delta T_f}{K_f} = \dfrac{0.30}{1.86} = 0.161$

According to Raoult's law

$\dfrac{p^0 - p}{p^0} = \dfrac{\text{No. of moles of solute}}{\text{No. of moles of solvent}}$

$\dfrac{23.51-p}{23.51} = \dfrac{0.161}{0.161+1000/18}$

$\left(\because \text{No. of moles of } H_2O = \dfrac{1000}{18}\right)$

On usual calculations,

$$\frac{23.51-p}{23.51}=0.00289$$

$$p=23.51-23.51\times0.00289=23.51-0.068$$

$$p=23.44 \text{ mm Hg}$$

58. 00125.09

Sol. Volume of the surface = area × thickness

$$80\,cm^2\times\frac{0.005}{10}cm=\frac{1}{25}cm^3$$

Mass of Ag deposited = Volume × Density

$$=\frac{1}{25}\times10.5\,g/cm^3=\frac{21}{50}g$$

Cell reaction: $Ag^+ + e^- \rightarrow Ag$

We know that, $\frac{W}{E}=\frac{Q}{F}=\frac{it}{F}$

E = Eq. wt. of Ag = 108

$$\therefore \frac{21/50}{108}=\frac{i\times t}{96500}$$

$$\frac{21}{50\times108}=\frac{3\times t}{96500}\therefore t=125.09\,sec$$

59. 00065.72

Sol. According to Arrhenius equation

$$\log\frac{k_2}{k_1}=\frac{E_a}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

Calculation of k at $25^oC(298K)$, i.e. k_1

Here $a=100, a-x=100-25, t=20\,min$

Thus $k_1=\frac{2.303}{20}\log\frac{100}{75}=0.01382\,min^{-1}$

Calculation of k at 40^oC (313 K) i.e., k_2

Substituting various values, we get

$$\log\frac{k_2}{0.01382}=\frac{70\times10^3}{2.303\times8.314}\times\left(\frac{1}{298}-\frac{1}{313}\right)$$

$$=\frac{70\times10^3\times15}{2.303\times8.314\times298\times313}=0.5879$$

$$k_2=0.01382\times3.872; k_2=0.0535\,min^{-1}$$

Calculation of % decomposition at 40^oC

Thus here a = 100, a – x = 100 – x, t = 20 min., k_2 = 0.0535 min^{-1}

Substituting the values in the first order reaction equation:

$$k_2=0.0535=\frac{2.303}{20}\log\frac{100}{100-x}$$

On usual calculation, x = 65.72%

60. 00059.33

Sol. $3MnO_2 \longrightarrow Mn_3O_4 + O_2$

$3\times(54.9+32)$ $(3\times54.9+64)$

$=260.7g$ $=228.7g$

Let the amount of pyrolusite ignited = 100.00 g

$\therefore$ Wt of MnO_2 = 80 g (80% of 100 g = 80)

Wt. of SiO_2 and other inert substances = 15 g

Wt of water = 100 – (80 + 15) = 5g

According to equation,

260.7 g of MnO_2 gives = 228.7 g of Mn_3O_4

$\therefore$ 80 g of MnO_2 gives = $\frac{228.7}{260.7}\times 80 = 70.18\,g$ of Mn_3O_4

During ignition, H_2O present in pyrolusite is removed while silica and other inert substances remain as such.

$\therefore$ Total wt. of the residue = 70.18 + 15 = 85.18 g

Calculation of % of Mn in ignited Mn_3O_4

3 Mn = Mn_3O_4

$3\times 54.9 = 164.7\,g$ $3\times 54.9 + 64 = 228.7g$

Since, 228.7 g of Mn_3O_4 contains 164.7 g of Mn

70.18 g of Mn_3O_4 contains $= \frac{164.7}{228.7}\times 70.18 = 50.54g$ of Mn

Weight of residue = 85.18 g

Hence, percentage of Mn is the ignited sample $= \frac{50.54}{85.18}\times 100 = 59.33\%$

Mathematics

PART – C

SECTION – A

61. D

Sol. $\because \log_{1/3}\left\{\log_{1/2}\left(|z|^2+4|z|+3\right)\right\}<0$

$\Rightarrow \quad \log_{1/2}\left(|z|^2+4|z|+3\right)>\left(\frac{1}{3}\right)^0$

$\Rightarrow \log_{1/2}\left(|z|^2+(z)+3\right)>1$

$\Rightarrow \quad |z|^2+4|z|+3<\left(\frac{1}{2}\right)^1$

$\Rightarrow \quad |z|^2+4|z|+\frac{5}{2}<0$

Positive quantity < 0 (Impossible)
Hence, S be an empty set or null set

62. A

Sol. $\because |x^2-2x-3|=b$

Option (A): for b < 0

$|x^2-2x-3|<0$

Impossible, there are no solutions

Option (B): For b = 0

$x^2-2x-3=0$

$\therefore x=3,-1$, there are two solutions

Option (C): For 0 < b < 1

$0<|x^2-2x-3|<1$

$\Rightarrow \quad 0<x^2-2x-3<1$

and $\quad -1<x^2-2x-3<0$

$\Rightarrow \quad x\in\left(1-\sqrt{5},-1\right)\cup\left(3,1+\sqrt{5}\right)$

and $\quad x\in\left(-1,1-\sqrt{3}\right)\cup\left(1+\sqrt{3},3\right)$

Option (D): For b = 1

$|x^2-2x-3|=1$

$\Rightarrow \quad x^2-2x-3=\pm1$

$\therefore \quad x=1\pm\sqrt{5}$ and $x=1\pm\sqrt{3}$

There are four solutions.

63. A

Sol. If n is even, the greatest coefficient is ${}^nC_{n/2}$

$\therefore$ The greatest term $={}^nC_{n/2}x^{n/2}$

$\because \quad {}^nC_{n/2}x^{n/2}>{}^nC_{(n/2)-1}x^{(n/2)-1}$

and $\quad {}^nC_{n/2}x^{n/2}>{}^nC_{(n/2)+1}x^{(n/2)+1}$

$$\Rightarrow \quad \frac{n-\frac{n}{2}+1}{\frac{n}{2}}x>1$$

and $$\frac{n-\left(\frac{n}{2}+1\right)+1}{\frac{n}{2}+1}x<1$$

$$\Rightarrow \quad x>\frac{n/2}{(n/2)+1} \text{ and } x<\frac{\frac{n}{2}+1}{\frac{n}{2}}$$

$$\Rightarrow \quad x>\frac{n}{n+2} \text{ and } x<\frac{n+2}{n}$$

Hence, $\frac{n}{n+2}<x<\frac{n+2}{2}$

64. B

Sol. $\because f(x)-x=0$ has imaginary roots,

then $f(x)-x>0$ or $f(x)-x<0, \forall x \in R$

for $f(x)-x>0, \forall x \in R$

then $f\{f(x)\}-f(x)>0, \forall x \in R$

Adding we get, $f\{f(x)\}-x>0, \forall x \in R$

Similarly, $f\{f(x)\}-x<0, \forall x \in R$

Thus, roots of equation $f\{f(x)\}-x=0$ are imaginary.

Let $$z=\begin{vmatrix}2 & \alpha & \delta\\ \beta & 0 & \alpha\\ \gamma & \beta & 1\end{vmatrix}$$

Then $$\bar{z}=\begin{vmatrix}2 & \bar{\alpha} & \bar{\delta}\\ \bar{\beta} & 0 & \bar{\alpha}\\ \bar{\gamma} & \bar{\beta} & 1\end{vmatrix}=\begin{vmatrix}2 & \beta & \gamma\\ \alpha & 0 & \beta\\ \delta & \alpha & 1\end{vmatrix}=\begin{vmatrix}2 & \alpha & \delta\\ \beta & 0 & \alpha\\ \gamma & \beta & 1\end{vmatrix}=z$$

Hence, z is purely real.

65. C

Sol. $(x)=\sin^{-1}[\sec x]$

f(x) is defined when

$-1 \le [\sec x] \le 1$

$\because \quad [\sec x] \le 1$

$\Rightarrow \quad \sec x < 2$

But $\sec x \ge 1$

$\therefore \quad 1 \le \sec x < 2$

$\Rightarrow \quad x \in \left[2m\pi, 2m\pi + \frac{\pi}{3}\right)$ ...(i)

$m \in I$

and if $[\sec x] \geq -1$

$\therefore \quad \sec x \geq -1$

But $\sec x \geq 1$

and $\sec x \leq -1$

$\therefore \quad \sec x \leq -1$

$\sec x = -1$

$x = 2n\pi + \pi$

$= (2n+1)\pi, n \in I$ ...(ii)

$\therefore$ From Eqs. (i) and (ii), we get

$D_f = \left\{(2n+1)\pi, n \in I\right\} \cup \left\{\left[2m\pi, 2m\pi + \frac{\pi}{3}\right), m \in I\right\}$

66. A

Sol. $\because 0 < \cos^2(n!\pi x) \leq 1$

If $0 < \cos^2(n!\pi x) < 1$

Then, $f(x) = \lim\limits_{m\to\infty} \lim\limits_{n\to\infty} \left\{1 + \left(\cos^2(n!\pi x)\right)^m\right\}$

$= 1 + 0 = 1$

When x is irrational

and if $\cos^2(n!\pi x) = 1$

$\Rightarrow \quad n!\pi x = r\pi \Rightarrow x = \frac{r}{n!} = \text{rational}$

Then, $f(x) = \lim\limits_{m\to\infty} \lim\limits_{m\to\infty} \left\{1 + \left(\cos^2 n!\pi x\right)^m\right\}$

$= 1 + 1 = 2$

Hence, $f(x) = \begin{cases} 2, x \text{ is rational} \\ 1, x \text{ is irrational} \end{cases}$

67. D

Sol. $\because \quad f(x) = \lim\limits_{n\to\infty} \frac{x^n - x^{-n}}{x^n + x^{-n}}, x > 1$

$= \lim\limits_{n\to\infty} \frac{1 - x^{-2n}}{1 + x^{-2n}}$

$= \frac{1-0}{1+0} = 1 \qquad (\because x > 1, \therefore x^{-\infty} = 0)$

Then, $\int \frac{xf(x)\ln\left(x + \sqrt{(1+x^2)}\right)}{\sqrt{(1+x^2)}} dx$

$= \int \frac{x\ln\left(x + \sqrt{(1+x^2)}\right)}{\sqrt{(1+x^2)}} dx$

Put $\ln\left(x+\sqrt{\left(1+x^2\right)}\right)=t \Rightarrow x+\sqrt{\left(1+x^2\right)}=e^t$

$\therefore \quad \dfrac{1+\dfrac{x}{\sqrt{\left(1+x^2\right)}}}{\left(x+\sqrt{1+x^2}\right)}dx=dt$

$\Rightarrow \quad \dfrac{dx}{\sqrt{\left(1+x^2\right)}}=dt$

and $\quad \left(e^t-x\right)^2=1+x^2$

$\Rightarrow \quad e^{2t}-2x\ e^t=1$

or $\quad x=\dfrac{1}{2}\left(e^t-e^{-t}\right)$

$\therefore \quad \int\dfrac{x\ln\left(x+\sqrt{\left(1+x^2\right)}\right)}{\sqrt{\left(1+x^2\right)}}dx$

$=\int\dfrac{1}{2}\left(e^t-e^{-t}\right).t\ dt$

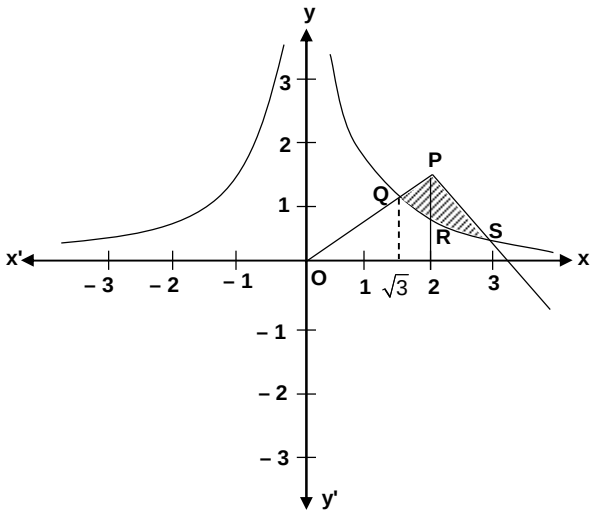
$=\dfrac{1}{2}\left\{t\left(e^t+e^{-1}\right)-\left(e^t-e^{-t}\right)\right\}+c$

$=\sqrt{\left(1+x^2\right)}.\ln\left(x+\sqrt{\left(1+x^2\right)}\right)-x+c$

68. C

Sol. $y=2-|2-x|, y=\dfrac{3}{|x|}$

$y=\begin{cases}x, & x\le 2\\ 4-x, & x\ge 2\end{cases}; y=\begin{cases}\dfrac{3}{x}, & x>0\\ -\dfrac{3}{x}, & x<0\end{cases}$



Hence, required area

$$PQRSP = \text{area } PQRP + \text{area } PRSP$$

$$= \left|\int_{\sqrt{3}}^{2}\left(x-\frac{3}{x}\right)dx\right| + \left|\int_{2}^{3}\left((4-x)-\frac{3}{x}\right)dx\right|$$

$$= \frac{4-3\ln 3}{2} \text{ sq unit.}$$

69. B

Sol. $\because \quad \sin y\frac{dy}{dx} = \cos y(1-x\cos y)$

$$\Rightarrow \quad \tan y\frac{dy}{dx} = 1-x\cos y$$

$$\Rightarrow \quad \sec y\tan y\frac{dy}{dx} - \sec y = -x \qquad \ldots(i)$$

Put $\sec y = v$

$$\therefore \quad \sec y\tan y\frac{dy}{dx} = \frac{dv}{dx}$$

Then, from Eq. (i), $\frac{dv}{dx} - v = -x$

$$\therefore \quad IF = e^{\int -1dx} = e^{-x}$$

Then, $v.\left(e^{-x}\right) = \int(-x)e^{-x}dx$

$$\Rightarrow \quad v.e^{-x} = (-x)\left(-e^{-x}\right) + e^{-x} + c$$

or $\quad v = x+1+ce^{x}$

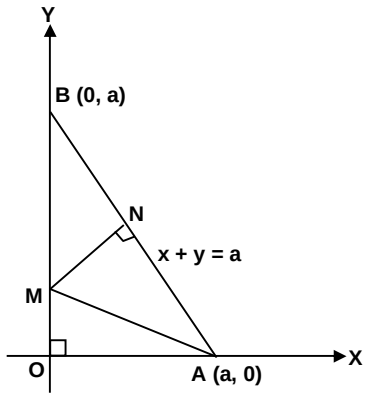
or $\quad \sec y = x+1+ce^{x}$

70. C

Sol. Let $\frac{AN}{BN} = \lambda$

Then, coordinate of N are $\left(\frac{a}{1+\lambda}, \frac{a\lambda}{1+\lambda}\right)$

$\because$ Slope of AB = –1



$\therefore$ Slope of MN = 1

$\therefore$ Equation on MN is

$$y - \frac{a\lambda}{1+\lambda} = x - \frac{a}{1+\lambda} \Rightarrow x - y = a\left(\frac{1-\lambda}{\lambda+1}\right)$$

So, the coordinates of M are $\left(0,\ a\left(\frac{\lambda-1}{\lambda+1}\right)\right)$

Therefore, area of $\Delta AMN = \frac{3}{8}$ area of ΔAOB

$\Rightarrow \quad \frac{1}{2}.AN.MN = \frac{3}{8}.\frac{1}{2}a.a$

$\Rightarrow \quad \frac{1}{2}.\left|\frac{a\lambda\sqrt{2}}{1+\lambda}.\frac{a\sqrt{2}}{1+\lambda}\right| = \frac{3}{8}.\frac{1}{2}a.a$

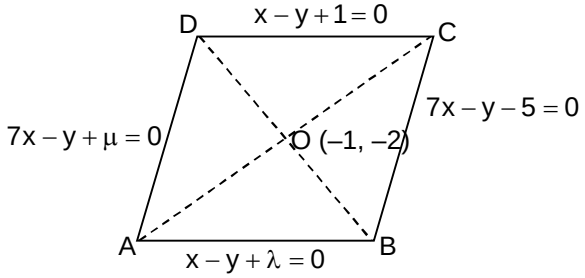
$\Rightarrow \quad \frac{a^2\lambda}{(1+\lambda)^2} = \frac{3}{8}.\frac{1}{2}a^2$

$\therefore \quad \lambda = 3$ or $\lambda = \frac{1}{3}$

For $\lambda = \frac{1}{3}$, then M lies outside the segment OB and hence the required value of $\lambda = 3$.

71. A

Sol.



Let other two sides of rhombus are $x - y + \lambda = 0$ and $7x - y + \mu = 0$

Then O is equidistant from AB and DC and from AD and BC.

$\therefore |-1+2+1| = |-1+2+\lambda| \Rightarrow \lambda = -3$

and $|-7+2-5| = |-7+2+\mu| \Rightarrow \mu = 15$

$\therefore$ Other two sides are $x - y - 3 = 0$ and $7x - y + 15 = 0$

On solving the equation of sides pairwise, we get the vertices as $\left(\frac{1}{3},\ \frac{-8}{3}\right)$, $(1,\ 2)$, $\left(\frac{-7}{3},\ \frac{-4}{3}\right)$, $(-3,\ -6)$.

72. D

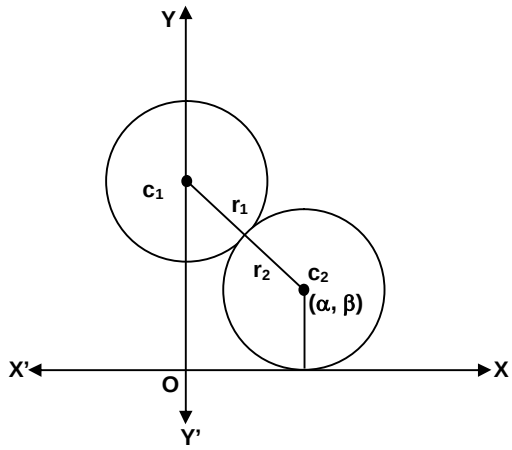
Sol. Equation of circle with centre (0, 3) and radius 2 is

$$x^2 + (y-3)^2 = 4$$

Let centre of the variable circle is (α,β)

$\because$ It touches X-axis

$\because$ It's equation is $(x-\alpha)^2 + (y+\beta)^2 = \beta^2$



Circle touch externally

$\Rightarrow \quad c_1c_2 = r_1 + r_2$

$\therefore \quad \sqrt{\alpha^2 + (\beta - 3)^2} = 2 + \beta$

$\alpha^2 + (\beta - 3)^2 = \beta^2 + 4 + 4\beta$

$\Rightarrow \quad \alpha^2 = 10(\beta - 1/2)$

$\therefore$ Locus is $x^2 = 10\left(y - \frac{1}{2}\right)$ which is a parabola.

73. A

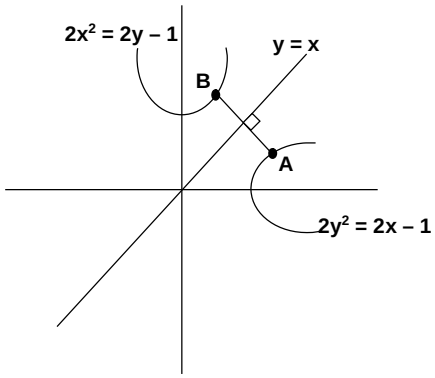
Sol. The given parabolas $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$ are symmetrical about the line y = x. The shortest distance occurs along the common normal which is perpendicular to the line y = x.

Therefore, the tangent at point A on $2y^2 = 2x - 1$ is parallel to y = x. therefore,

$$4y\frac{dy}{dx} = 2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{2y} = 1$$

or $y = \frac{1}{2}$ and $x = \frac{3}{4}$

$\therefore \quad A \equiv \left(\frac{3}{4}, \frac{1}{2}\right)$, then coordinates of B on $2x^2 = 2y - 1$ is $B \equiv \left(\frac{1}{2}, \frac{3}{4}\right)$



$\therefore$ Shortest distance = AB

$$= \sqrt{\left(\frac{3}{4}-\frac{1}{2}\right)^2+\left(\frac{1}{2}-\frac{3}{4}\right)^2}$$

$$= \sqrt{\left(\frac{1}{16}+\frac{1}{16}\right)} = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

74. D

Sol. Equation of tangent of the ellipse at $P(5\cos\theta, 4\sin\theta)$ is

$$\frac{x}{5}\cos\theta + \frac{y}{4}\sin\theta = 1$$

It meets the line x = 0 at Q $(0,\ 4\operatorname{cosec}\theta)$ and image of Q in the line y = x is $R(4\operatorname{cosec}\theta, 0)$

$\therefore$ Equation of the circle is

$(x-0)(x-4\operatorname{cosec}\theta)+(y-4\operatorname{cosec}\theta)(y-0)=0$

i.e. $x^2+y^2-4(x+y)\operatorname{cosec}\theta = 0$

$\therefore$ Each member of the family passes through the intersection of

$x^2+y^2=0$ and $x+y=0$

i.e., then point (0, 0).

75. A

Sol. Equation of plane through $(2,-1,0)$ is

$a(x-2)+b(y+1)+c(z-0)=0$ ...(i)

It also passes through $(3,-4,5)$, then

$a-3b+5c=0$ …(ii)

Given plane (i) is parallel to a line which direction cosines proportional to 2, 3, 4

$\therefore$ $2a+3b+4c=0$ …(iii)

From Eqs. (ii) and (iii), we get

$$\frac{a}{9}=\frac{b}{-2}=\frac{c}{-3}=\lambda \text{ (say)}$$

$\therefore$ $a=9\lambda, b=-2\lambda, c=-3\lambda$

Now, from Eq. (i)

$9\lambda(x-2)-2\lambda(y+1)-3\lambda(z-0)=0$

$\Rightarrow$ $9x-2y-3z=20$

$\therefore$ $k=20$

76. A

Sol. $\because\ 0<\alpha<\frac{\pi}{6}$

$\Rightarrow$ $0<\frac{\alpha}{2}<\frac{\pi}{12}$ (in first quadrant),

then tan $\alpha/2$ is always positive

$\because$ $\sin\alpha+\cos\alpha=\frac{\sqrt{7}}{2}$

$\Rightarrow$ $$\frac{2\tan\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}}+\frac{1-\tan^2\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}}=\frac{\sqrt{7}}{2}$$

$\Rightarrow \quad 2\left(2\tan\frac{\alpha}{2}+1-\tan^2\frac{\alpha}{2}\right)=\sqrt{7}\left(1+\tan^2\frac{\alpha}{2}\right)$

$\Rightarrow \quad \left(\sqrt{7}+2\right)\tan^2\frac{\alpha}{2}-4\tan\frac{\alpha}{2}+\left(\sqrt{7}-2\right)=0$

$\therefore \quad \tan\frac{\alpha}{2}=\frac{4\pm\sqrt{16-4\left(\sqrt{7}+2\right)\left(\sqrt{7}-2\right)}}{2\left(\sqrt{7}-2\right)}$

$=\frac{4\pm\sqrt{16-12}}{2\left(\sqrt{7}+2\right)}$

$=\frac{4\pm2}{2\left(\sqrt{7}+2\right)}=\frac{2\pm1}{\left(\sqrt{7}+2\right)}$

$=\frac{1}{\sqrt{7}+2}\text{ or }\frac{3}{\sqrt{7}+2}$

$=\frac{\sqrt{7}-2}{3}\text{ or }\sqrt{7}-2$

Hence, $\tan(\alpha/2)=\frac{\sqrt{7}-2}{3}$ $\qquad\left(\because 0<\left(\frac{\alpha}{2}\right)<\frac{\pi}{12}\right)$

77. A

Sol. $\sin^{10}x+\cos^{10}x=\frac{29}{16}\cos^4 2x$

$\left(\frac{1-\cos2x}{2}\right)^5+\left(\frac{1+\cos2x}{2}\right)^5=\frac{29}{16}\cos^4 2x$

$\Rightarrow \quad \frac{1}{32}\times2\left\{1+10\cos^2 2x+5\cos^4 2x\right\}=\frac{29}{16}\cos^4 2x$

$\Rightarrow \quad 24\cos^4 2x-10\cos^2 2x-1=0$

$\Rightarrow \quad \left(2\cos^2 2x-1\right)\left(12\cos^2 2x+1\right)=0$

If $\quad \cos^2 2x=\frac{1}{2} \qquad \left(\because 12\cos^2 2x+1\neq0\right)$

$\Rightarrow \quad 2\cos^2 2x-1=0$

$\Rightarrow \quad \cos4x=0$

$4x=n\pi+\frac{\pi}{2}$

$\Rightarrow \quad x=\frac{n\pi}{4}+\frac{\pi}{8},n\in I$

78. B

Sol. Since, $\frac{a\cos A+b\cos B+c\cos C}{a\sin B+b\sin C+c\sin A}=\frac{a+b+c}{9R}$

$\Rightarrow \quad \frac{2R\sin A\cos A+2R\sin B\cos B+2R\sin C\cos C}{a\times\frac{b}{2R}+b\times\frac{c}{2R}+c\times\frac{a}{2R}}=\frac{a+b+c}{9R}$

$$\Rightarrow \quad \frac{R(\sin 2A+\sin 2B+\sin 2C)}{\frac{(ab+bc+ca)}{2R}}=\frac{a+b+c}{9R}$$

$$\Rightarrow \quad \frac{R(4\sin A\sin B\sin C).2R}{(ab+bc+ca)}=\frac{a+b+c}{9R}$$

$$\Rightarrow \quad \frac{2R^2\left(4.\frac{a}{2R}.\frac{b}{2R}.\frac{c}{2R}\right)}{(ab+bc+ca)}=\frac{a+b+c}{9R}$$

$$9abc=(a+b+c)(ab+bc+ca)$$

$$\Rightarrow \quad a(b-c)^2+b(c-a)^2+c(a-b)^2=0$$

Which is possible only when

$$a-b=0, b-c=0, c-a=0$$

$$\therefore \quad a=b=c$$

79. B

Sol. Given, $\vec{r}.\vec{a}=0$

and $\left|\vec{r}\times\vec{b}\right|=|\vec{r}||\vec{b}|$

Let angle between $\vec{r}$ and $\vec{b}$ is θ

$$\therefore \quad =\left|\vec{r}\times\vec{b}\right|=|\vec{r}||\vec{b}|\sin\theta=|\vec{r}||\vec{b}|$$

$$\therefore \quad \theta=\frac{\pi}{2}$$

and let angle between $\vec{r}\times\vec{c}$ is ϕ,

Then, $|\vec{r}\times\vec{c}|=|\vec{r}||\vec{c}|\sin\phi=|\vec{r}||\vec{c}|$

$\therefore$ are coplanar = $\frac{\pi}{2}$

$\therefore \vec{r}\perp\vec{a}$, $\vec{r}\perp\vec{b}$ and $\vec{r}\perp\vec{c}$

Hence, $\vec{a}$, $\vec{b}$ and $\vec{c}$

$$\therefore \quad \left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$$

80. B

Sol. Let $\overrightarrow{OA}=\vec{a}, \overrightarrow{OB}=\vec{b}$ and $\overrightarrow{OC}=\vec{c}$, then

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\vec{b}-\vec{a}$$

and $\overrightarrow{OP}=\frac{1}{3}\vec{a}$, $\overrightarrow{OQ}=\frac{1}{2}\vec{b}, \overrightarrow{OR}=\frac{1}{3}\vec{c}$

and $\overrightarrow{OS}=\lambda\left(\vec{b}-\vec{a}\right)$

$\because$ P,Q,R and S are coplanar, then

$\overrightarrow{PS}=\alpha\overrightarrow{PQ}+\beta\overrightarrow{PR}$ ($\overrightarrow{PS}$ can be written as a linear combination of $\overrightarrow{PQ}$ and $\overrightarrow{PR}$)

$$\overrightarrow{OS}-\overrightarrow{OP}=\alpha\left(\overrightarrow{OQ}-\overrightarrow{OP}\right)+\beta\left(\overrightarrow{OR}-\overrightarrow{OP}\right)$$

$$\Rightarrow \quad \overrightarrow{OS}=\alpha\overrightarrow{OQ}+\beta\overrightarrow{OR}+(1-\alpha-\beta)\overrightarrow{OP}$$

$$\Rightarrow \quad \lambda\left(\vec{b}-\vec{a}\right)=\frac{\alpha\vec{b}}{2}+\beta\frac{\vec{c}}{3}+(1-\alpha-\beta)\frac{\vec{a}}{3}$$

On comparing

$$-\lambda = \frac{(1-\beta)}{3}, \lambda = \frac{\alpha}{2}$$

and $\beta = 0$

$\therefore$ $-\lambda = \frac{1-\alpha}{3}, \lambda = \frac{\alpha}{2}$

$\Rightarrow$ $-3\lambda = 1 - 2\lambda$

$\therefore$ $\lambda = -1$

SECTION – B

81. 00015.00

Sol. $y = 2[x] + 3 = 3[x-2] + 5$

$\Rightarrow$ $y = 2[x] + 3 = 3[x] - 6 + 5$

$\Rightarrow$ $2[x] + 3 = 3[x] - 1$

$\Rightarrow$ $[x] = 4, \therefore\ y = 11$

Then $[x + y] = y + [x] = 11 + 4 = 15$

82. 00013.00

Sol. We have

$$(x-1)(x^2-2)(x^3-3)(x^4-4).....(x^{20}-20) \quad ...(i)$$

$$= x^{190}(x-1)\left(x-\frac{2}{x}\right)\left(x-\frac{3}{x^2}\right)\left(x-\frac{4}{x^3}\right)...\left(x-\frac{20}{x^{19}}\right)$$

$$= x^{190}\left\{\begin{array}{l} x^{20} - x^{19}\left(1+\frac{2}{x}+\frac{3}{x^2}+\frac{4}{x^3}+\frac{5}{x^4}+\frac{6}{x^5}+\frac{7}{x^6}+...\right) \\ \quad +x^{18}\left(\left(\frac{2}{x}+\frac{3}{x^2}+\frac{4}{x^3}+\frac{5}{x^4}+\frac{6}{x^5}+...\right)\right. \\ \quad \left.+\left(\frac{6}{x^3}+\frac{8}{x^4}+\frac{10}{x^5}+...\right)+\frac{12}{x^5}+...\right) \\ \quad -x^{17}\left(\frac{6}{x^3}+\frac{8}{x^4}+...\right)+x^{16}(.....) \end{array}\right\}$$

$\therefore$ Coefficient of x^{203} in Eq. (i)

$$= -7 + 6 + 10 + 12 - 8 = 13$$

83. 00536.00

Sol. Coefficient of $x^{20}\ln(1-x)^{-3}(1-x^4)^{-1}$

Coefficient of x^{20} in

$$\left(1 + {}^3c_1 x + {}^4c_2 x^2 + ...\right)\left(1 + x^4 + x^4 + x^{12} + x^{16} + x^{20}\right)$$

$$= 1 + {}^6c_4 + {}^{10}c_8 + {}^{14}c_{12} + {}^{18}c_{16} + {}^{22}c_{20}$$

$$= 1 + {}^6c_2 + {}^{10}c_2 + {}^{14}c_2 + {}^{18}c_2 + {}^{22}c_2$$

$$= 1 + 15 + 45 + 45 + 91 + 153 + 231$$

$$= 536$$

84. 00896.00

Sol. $\because \quad \sum_{r=0}^{7} A_r = \sum_{r=0}^{7} \begin{pmatrix} a_r & 0 \\ 0 & b_r \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$

$\therefore \quad a = \sum_{r=0}^{7} a_r = \sum_{r=0}^{7} r.\left({}^7C_r\right)$

and $\quad b = \sum_{r=0}^{7} b_r = \sum_{r=0}^{7} (7-r).\left({}^7C_r\right)$

$\Rightarrow \quad a + b = 7\sum_{r=0}^{7} {}^7C_r = 7\left(2^7\right) = 7 \times 128 = 896$

85. 00144.00

Sol. Let A denote the event that the target is hit when x shells are fired at point I. Let $E_1(E_2)$ denote the event,

We have, $P(E_1) = \frac{8}{9}, P(E_2) = \frac{1}{9}$

$\Rightarrow \quad P(A/E_1) = 1 - \left(\frac{1}{2}\right)^x$ and $P(A/E_2) = 1 - \left(\frac{1}{2}\right)^{21-x}$

Now, $\quad P(A) = \frac{8}{9}\left[1 - \left(\frac{1}{2}\right)^x\right] + \frac{1}{9}\left[1 - \left(\frac{1}{2}\right)^{21-x}\right]$

$\Rightarrow \quad \frac{dP(A)}{dx} = \frac{8}{9}\left[\left(\frac{1}{2}\right)^x \log 2\right] + \frac{1}{9}\left[-\left(\frac{1}{2}\right)^{21-x} \log 2\right]$

Now, we must have $\frac{dP(A)}{dx} = 0$

$\Rightarrow \quad x = 12, \text{also} \frac{d^2P(A)}{dx^2} < 0$

Hence, P(A) is maximum when x = 12.

$\therefore \quad 12x = 12 \times 12 = 144$

86. 04824.00

Sol. The period of $\sin\left(\frac{\pi[x]}{12}\right)$ is $\frac{2\pi}{\frac{\pi}{12}} = 24$

as $\sin\left(\frac{\pi[x+24]}{12}\right) = \sin\left(\frac{\pi([x]+24)}{12}\right)$

$= \sin\left(2\pi + \frac{\pi[x]}{12}\right) = \sin\left(\frac{\pi[x]}{12}\right)$

similarly the period of $\tan\left(\frac{\pi[x]}{3}\right)$ is 3 and the period of $\cos\left(\frac{\pi x}{4}\right)$ is $\frac{2\pi}{\frac{\pi}{4}} = 8$

Hence, the period of the given function

$\lambda = \text{LCM of } 24, 8, 3 = 24$

$\therefore 201\lambda = 201 \times 24 = 4824$

87. 02500.00

Sol. Given, $f(x)+f(y)=f\left(\dfrac{x+y}{1-xy}\right)$

Putting x = 0, y = 0, we get f(0) = 0 …(i)

and putting $y=-x$, we get

$f(x)+f(-x)=f(0)=0$

$\therefore \quad f(x)=-f(-x)$ ...(ii)

Now, $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$=\lim\limits_{h\to 0}\dfrac{f(x+h)+f(-x)}{h}$ [from Eq.(ii)]

$=\lim\limits_{h\to 0}\dfrac{f\left(\dfrac{x+h-x}{1+(x+h)x}\right)}{h}$

$=\lim\limits_{h\to 0}\dfrac{f\left(\dfrac{h}{1+x^2+xh}\right)}{\left(\dfrac{h}{1+x^2+xh}\right).(1+x^2+xh)}$

$f'(x)=\dfrac{2}{1+x^2}$ $\left(\because \lim\limits_{x\to 0}\dfrac{f(x)}{x}=2\right)$

$\therefore \quad f'(-2)=\dfrac{2}{1+4}=\dfrac{2}{5}$

and $f(x)=2\tan^{-1}x+c$

$f(0)=0+c=0$

$\therefore \quad c=0$

$f(x)=2\tan^{-1}x$

$\Rightarrow \quad f\left(\sqrt{3}\right)=\tan^{-1}\sqrt{3}=\dfrac{2\pi}{3}$

$\therefore \quad \dfrac{1500}{\pi}.\dfrac{f\left(\sqrt{3}\right)}{f'(-2)}=\dfrac{1500}{\pi}.\dfrac{\frac{2\pi}{3}}{\frac{2}{5}}=\dfrac{1500}{\pi}.\dfrac{5\pi}{3}=2500$

88. 02008.00

Sol. Let $f(x)=(e^x-1)-(1+x)\ln(1+x)$

$\therefore \quad f'(x)=e^x-\{1+\ln(1+x)\}=g(x)(\text{let})$

$\therefore \quad g(x)=e^x-1-\ln(1+x)$

$\Rightarrow \quad g'(x)=e^x-\dfrac{1}{(1+x)}>0 \quad (\text{for } x\in(0,\infty))$

$\Rightarrow \quad g(x)$ is increasing function

$\because \quad x > 0$

$\therefore \quad g(x) > g(0) \Rightarrow g(x) > 0$

Then, $f'(x) > 0$

$\Rightarrow$ $f(x)$ is increasing function

$f(x) > f(0)$

$\Rightarrow \quad e^x - 1 - (1+x)\ln(1+x) > 0$

$\Rightarrow \quad e^x - 1 > (1+x)\ln(1+x)$

$\therefore \quad \phi(x) = e^x - 1 \Rightarrow \phi(4) = e^4 - 1 \Rightarrow \phi(4) + 1 = e^4$

$\therefore \quad \ln(\phi(4)+1) = 4$

Then, $502\ln(\phi(4)+1) = 502 \times 4 = 2008$.

89. 01005.00

Sol. Let r be the radius of the circle and x be the side of the square.

Then, $\ell = 2\pi r + 4x$ ....(i)

Let S be the sum of the areas of the circle and the square, then

$$S = \pi r^2 + x^2$$

$$= \pi\left(\frac{\ell - 4x}{2\pi}\right)^2 + x^2 \qquad [\text{from Eq. (i)}]$$

$$= \frac{1}{4\pi}(\ell - 4x)^2 + x^2$$

$\therefore \quad \dfrac{dS}{dx} = \dfrac{2}{4\pi}(\ell - 4x)(-4) + 2x$

$$= -\frac{2}{\pi}(\ell - 4x) + 4x$$

and $\dfrac{d^2S}{dx^2} = \dfrac{8}{\pi} + 2 > 0$

$\therefore$ S is minimum

For max or min, $\dfrac{dS}{dx} = 0$

$\Rightarrow \quad -\dfrac{2}{\pi}(\ell - 4x) + 2x = 0$

or $x = \dfrac{\ell}{4+\pi}$ ...(ii)

From Eqs. (i) and (ii),

$$4x + \pi x = 2\pi r + 4x$$

$\Rightarrow \quad x = 2r$

$\therefore \quad \lambda = 1$

Then, $1005\lambda = 1005$

90. 00114.00

Sol. We have

$$\int_1^{xy} f(t)\,dt = y\int_1^x f(t)dt + f\int_1^y f(t)dt$$

Differentiating both sides w.r.t x keeping y as constant

$$f(xy)y = yf(x) + \int_1^y f(t)dt$$

Putting x = 1, then

$$yf(y) = yf(1) + \int_1^y f(t)dt$$

$\Rightarrow$ $yf(y) = 3y + \int_1^y f(t)dt \qquad (\because f(t) = 3)$

Again, differentiating both sides w.r.t y, then

$$yf'(y) + f(y).1 = 3 + f(y)$$

$\Rightarrow$ $f'(y) = \dfrac{3}{y}$

$\therefore$ $f(y) = 3\ln y + c$

$\Rightarrow$ $f(1) = 3\ln 1 + c = 3$

$\therefore$ $c = 3$

$\Rightarrow$ $f(y) = 3\ln y + 3 = 3\ln(ey)$

Then, $f\left(e^{37}\right) = 3\ln\left(e.e^{37}\right) = 3\ln\left(e^{38}\right)$

$= 3 \times 38 \ln e = 114$